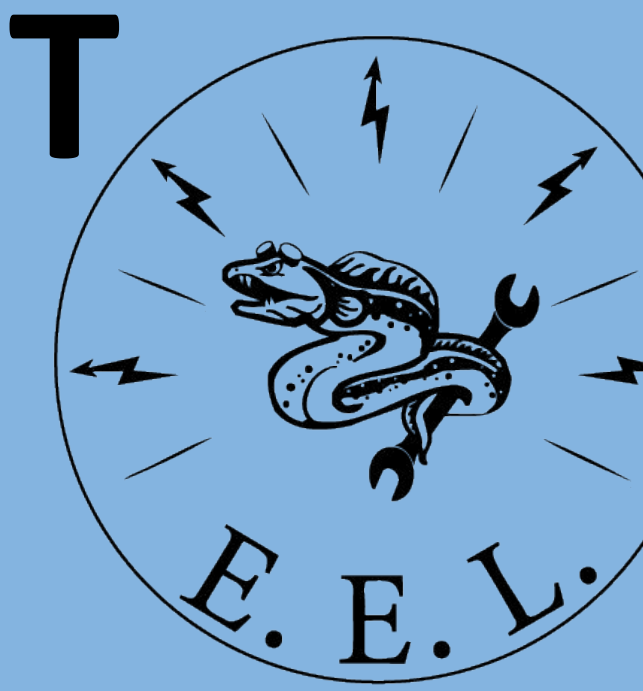


PAWS: Personal Autonomous Walking Support via Boston Dynamics SPOT

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Introduction

- This project presents an autonomous assistive navigation system using the Boston Dynamics SPOT robot to guide individuals with visual impairments
- The system integrates voice commands, autonomous navigation, and a physical guidance handle, removing the need for visual-based interaction.
- This system is designed for safe, accessible, and intuitive user guidance.

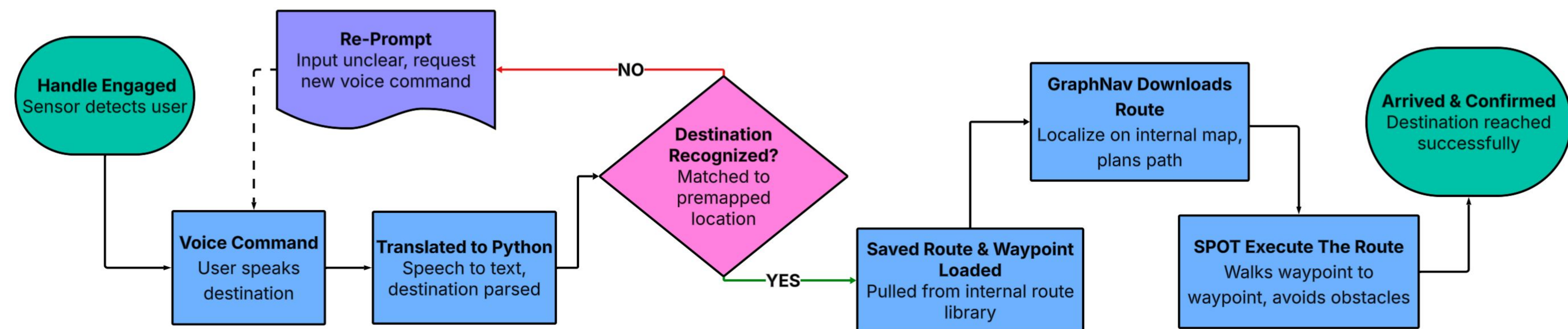


Fig. 1. This figure demonstrates how our guide dog walking support system will work internally. User input is processed, matched to predefined locations, and executed using SPOT’s GraphNav system, enabling waypoint-based navigation and obstacle avoidance. In addition to the components shown, the system includes a physical guidance handle that allows the user to safely follow the robot without relying on visual cues. The navigation framework is designed to support future integration with external routing services such as Google Maps API, allowing users to specify destinations more naturally and enabling route generation beyond pre-mapped waypoints.

Specialized Handle

- User holding the leash engages the handle, allowing SPOT to accept commands and execute routes.

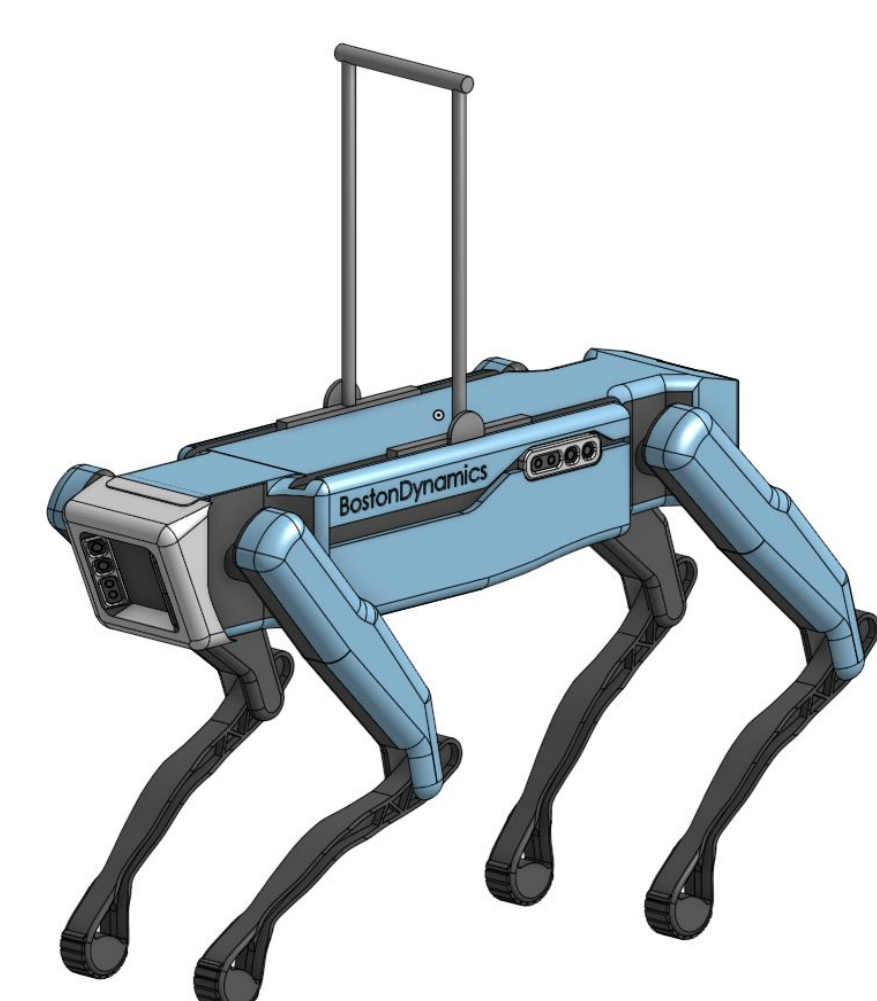


Fig. 2. Model of specialized handle integrated with SPOT.

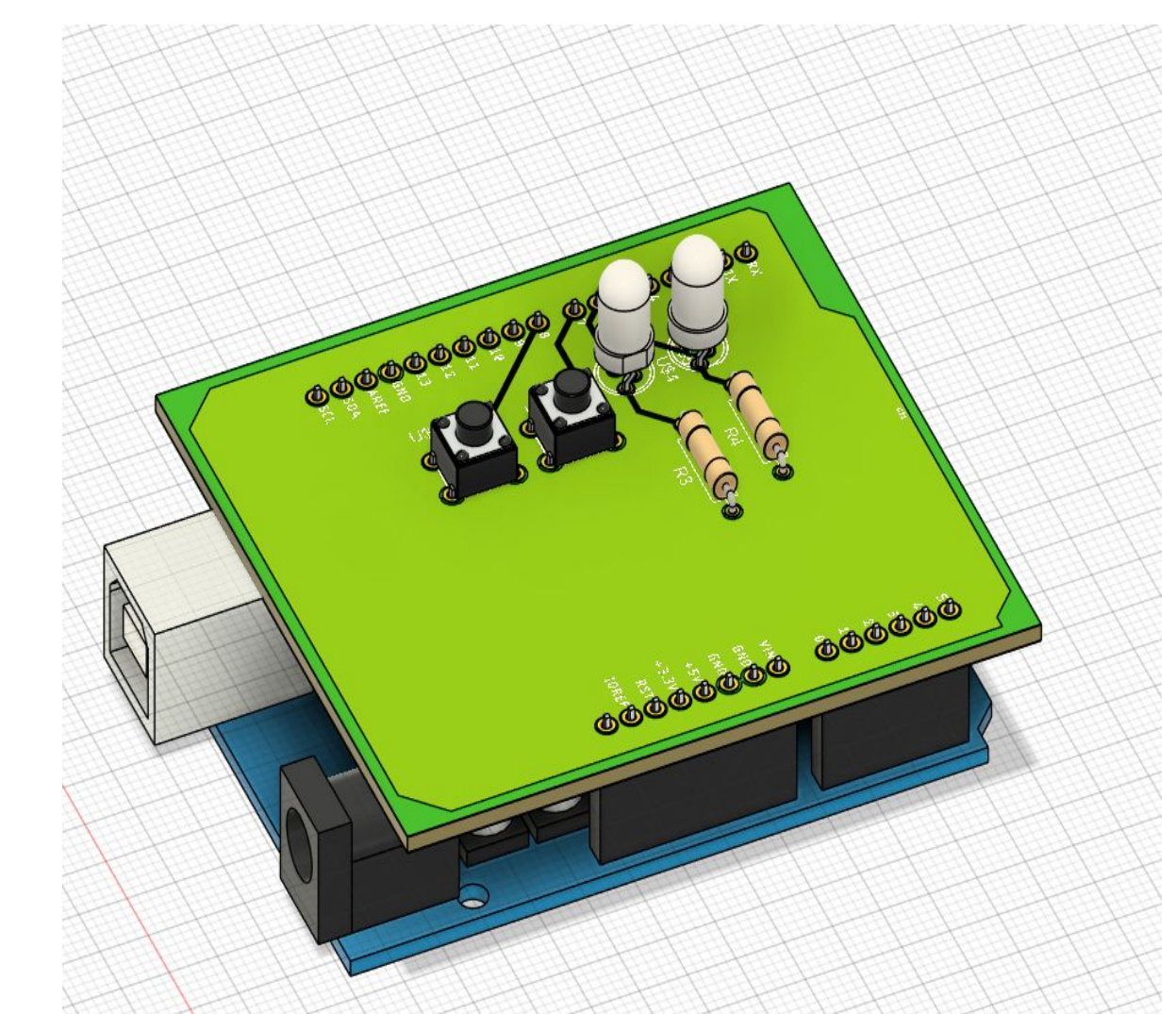


Fig. 3. Specialized handle circuit. One button engages the handle, and the other halts the route.

- Handle is released, SPOT immediately halts route execution and lies down, ensuring user safety in emergencies.

- The current handle uses a simple on/off circuit, with planned transition to capacitive touch sensing.

Voice Command

- Voice input is collected through a phone and run through a speech recognition tool to turn what the user says into text that the program can work with.
- A parser will then be created to match that text to a list of saved locations, also handling different ways people might phrase the same command or say something unclear.
- Once a destination is matched, the command will be sent to SPOT using its Python SDK.

Navigation

- A pre-mapped environment using GraphNav, represents locations as waypoints connected by navigable paths, enabling GPS-free localization.
- When a destination is requested, the system matches the command to a known waypoint and retrieves the corresponding route.
- LiDAR enables real-time obstacle detection and avoidance, allowing the robot to adapt to dynamic environments.
- Integration with the Google Maps API enables natural language destination requests, which translate into mapped routes for the robot to follow.

Conclusion

- This system combines navigation, voice interaction, and physical feedback to enable intuitive robotic guidance.
- Real-time localization, obstacle avoidance, and interrupt-based control ensure safe and responsive operation.
- Using the Boston Dynamics SPOT robot demonstrates the potential of adapting robots for assistive applications.

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Acknowledgements

We would like to thank the Experimental Engineering Laboratory for this opportunity and the endless support that made this possible.